INSTRUCTIONS

GEK-47613

STATOTROL* REGENERATIVE ADJUSTABLE SPEED DRIVE 6VHRS201A1

1.25 HP, SINGLE PHASE

230 VOLTS, 50/60 HERT

IMPORTANT

See Supplement on Page 15

DEPENDABLE MOTOR CONTROL

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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STATOTROL REGENERATIVE DC DRIVE

INTRODUCTION

This manual contains general information on the General Electric Statotrol regenerative D.C. Drive. Additional information and diagrams may be supplied with the equipment when necessary.

The regenerative Statotrol is a high performance, electronic reversing controller intended for use in applications requiring rapid acceleration/deceleration, closed loop positioning or high start-stop duty cycles.

DESCRIPTION

The Statotrol Drive consists of a controller and a D.C. motor. The controller contains forward and reverse SCRs, a loop contactor and the necessary electronic control circuitry. Built-in dynamic braking is also included.

The motor is a shunt wound DC motor specially designed for use with a half-wave phase controlled SCR power supply.

RECEIVING AND STORAGE

As soon as the equipment is received, it should be unpacked and examined for damage sustained in transit. If damage is evident, a damage claim should be filed immediately with the transportation company, and the local General Electric Co. Sales Office should be promptly notified.

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected against accidental damage. Avoid storage in a location where construction work is in progress.

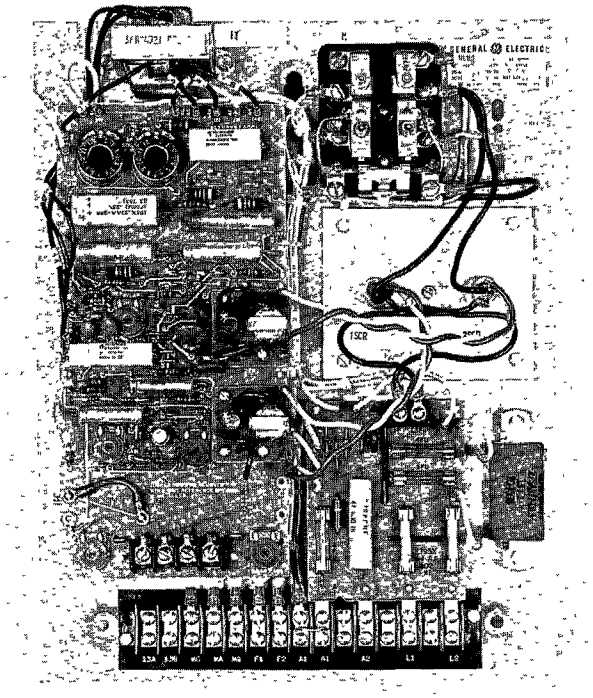


Figure 1. 6VHRS Regenerative Reversing Statotrol

INSTALLATION

CONTROLLER INSTALLATION

The Statotrol 6VHRS Control should be wall mounted in a location which will allow free flow of cooling air over the controller. Maximum ambient temperature around the controller must not exceed 40°F (104°F). Figure 2 provides mounting dimensions and outline dimensions. It is recommended that two or more inches of clearance be provided all around the controller to assure adequate air flow for cooling.

MOTOR INSTALLATION

The motor must be firmly mounted and properly aligned to prevent vibration. Excessive vibration causes rapid wear and audible noise. Heat dissipated by the motor will raise the ambient temperature around the motor if the motor is installed in an enclosed space. Since a lower ambient temperature will extend the motor life, it is always desirable to mount the motor in a well ventilated location. The motor should never be used where the ambient temperature exceeds 40°C (104°F) unless an oversized motor has been specially selected.

WARNING

IF THE MOTOR IS ACCESSABLE WHILE IT IS RUNNING, A PROTECTIVE GUARD SHOULD BE INSTALLED AROUND ALL EXPOSED ROTATING PARTS.

ELECTRICAL CONNECTIONS

PROPER LINE VOLTAGE CONNECTION

The controller operates on 50/60 cycle AC. Input voltage rating is 230 volts. A transformer is required where supply voltage deviates more than 10%.

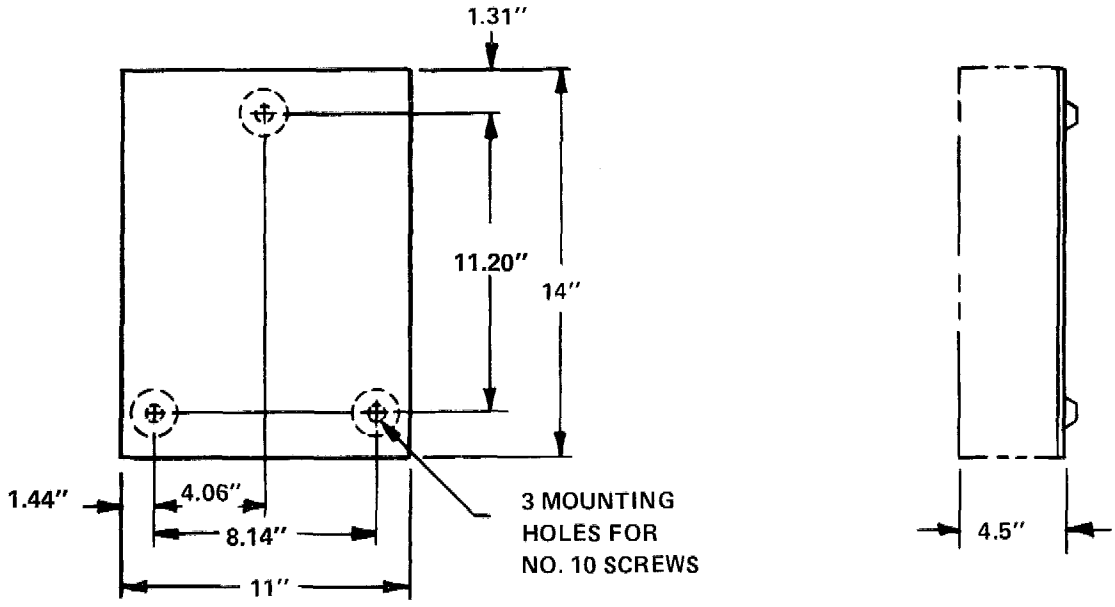


Figure 2. Control Outline and Mounting Dimensions

GROUNDING

WARNING

PERSONNEL SAFETY CONSIDERATIONS AND THE NATIONAL ELECTRIC CODE REQUIRE THAT ELECTRICAL APPARATUS ENCLOSURES BE SOLIDLY CONNECTED TO BUILDING GROUND.

The necessary wire gauge of the ground wire is determined by the rating of the branch circuit breaker. Connect a suitably grounded lead to the screw in the lower right hand corner of the SCR mounting plate. See Figure 4. Do not rely or depend on mounting bolts for a gound connection.

WARNING

PROPER MOTOR GROUNDING IS ESSENTIAL FOR PERSONNEL SAFETY.

Do not depend upon motor mounting bolts to ground the motor frame. The ground connection provided in the motor depends upon the type, size, and enclosure of the motor. Each Statotrol motor has one of the following provisions for connecting a ground lead.

1. A 4 inch green wire inside the wire well.
2. A 1/4-20 bolt in a tapped hole in the motor end shield.
3. A brass screw which serves as a ground terminal and also mounts the conduit box found on the side of TEFC motors.
4. A drilled hole in the motor end shield (inside the wire well near the electrical terminals) suitable for a self tapping screw.

If there is any doubt about motor lead identification, the problem can be resolved with a simple resistance check. The motor armature resistance will be about 0.3 to 15 ohms while field resistance will be about 75 to 100 ohms. There must be no continuity between armature and field. Permanent magnet DC motors have armature leads, but no field leads.

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. SINCE THE CONTROLLER STOP SWITCH AND FUSE DO NOT REMOVE VOLTAGE FROM THE CONTROLLER OR THE MOTOR FIELD, A BRANCH CIRCUIT BREAKER OR A FUSED DISCONNECT SWITCH MUST BE USED TO DISCONNECT THE CONTROLLER WHENEVER SERVICE WORK IS TO BE PERFORMED ON THE DRIVE.

POWER WIRING

A 15 point terminal board, 1TB, is provided for con-

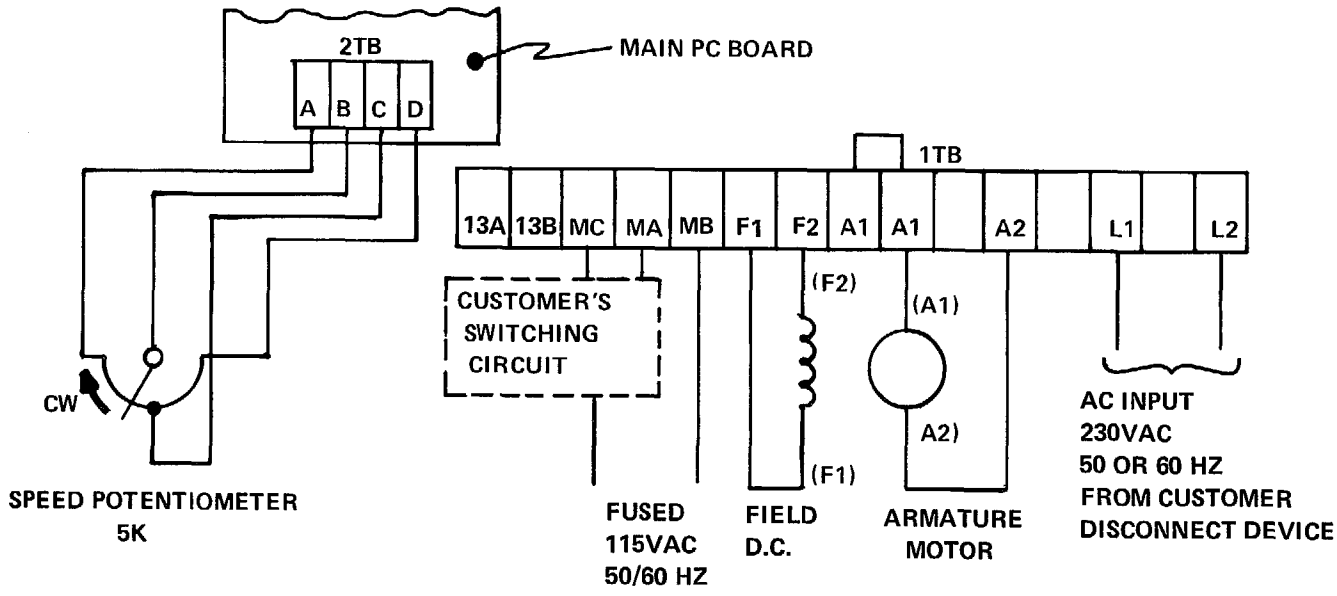


Figure 3. Customer Connection Diagram

nection to the AC power lines, motor leads, and control circuit. Reference (speed potentiometer) connections are made to the 4 point terminal board, 2TB on the main printed circuit board.

When the motor is connected as shown in Figure 3, a negative reference voltage (speed pot CCW) will result in CCW rotation of the motor viewed from the shaft end. Reversing the motor armature leads or the reference polarity will reverse motor rotation.

CAUTION

A CONNECTION ERROR IN THE POWER WIRING CAN DAMAGE THE CONTROL. RECHECK THE CONNECTIONS BEFORE POWER IS APPLIED.

CAUTION

TO PREVENT ELECTRICAL NOISE PICKUP, TWISTED OR SHIELDED WIRE IS RECOMMENDED IN ALL REFERENCE AND FEEDBACK CIRCUITS.

INITIAL OPERATION AND ADJUSTMENT

CURRENT LIMIT ADJUSTMENTS

The Statotrol current limit circuitry works by limiting the turn-on time of the SCR's. This limits the effective voltage across the motor armature, and the armature impedance then limits the current. The Statotrol controller is factory-set to limit the current in a low impedance motor.

SET POTENTIOMETER 3P

Turning potentiometer 3P clockwise will increase the current which will flow in the armature when the rotor is locked. Since motor output torque is proportional to armature current, 3P effectively adjusts the starting torque of the motor.

How to Set 3P by Measuring Armature Current

Connect a DC ammeter in series with the motor armature, lock the rotor, turn the reference potentiometer in either direction, and energize the M relay. Turn potentiometer 3P until the desired armature current is achieved. Do not stall the motor for more than one (1) minute or the motor may be damaged.

CAUTION

DO NOT SET DC STALL CURRENT TO EXCEED 200% OF MOTOR NAMEPLATE-RATED CURRENT OR DAMAGE MAY RESULT. CONSULT THE FACTORY IF HIGHER TORQUE IS REQUIRED.

How to Set 3P Without an Ammeter

An approximate setting for stall current can be achieved by aligning the correct ink spots on potentiometer 3P. The approximate correct setting is with the center dot aligned with the adjustment dot.

SET 4P

Turning 4P clockwise will increase the available motor output torque at rated motor speed.

How to Set 4P by Measuring Armature Current

With a DC ammeter in the armature circuit, lock the rotor, turn the reference potentiometer in either direction, and energize the M relay. Read the current in the armature, release the rotor, and observe the ammeter as the motor accelerates. Potentiometer 4P must be set so that the armature current holds steady or gradually decreases as the motor accelerates to top speed. Turning 4P clockwise will increase the armature current available at higher speeds. For smooth acceleration and long motor life, it is recommended that 4P be turned to the farthest counterclockwise setting which will allow desired motor performance.

How to Set 4P Without an Ammeter

An approximate setting for current limit can be achieved by aligning the correct ink spots on potentiometer 4P. The approximate correct setting is with the center dot aligned with the adjustment dot.

OVERLAP SETTING

Potentiometer 2P is the overlap adjustment. Turning 2P clockwise gives faster "breakaway" response and decreases the "deadband" as the reference signal goes from forward to reverse. Turning 2P clockwise also increases motor heating at low speed and may contribute to stability problems in certain applications. Therefore, 2P is set fully counterclockwise as a factory adjustment, and it is recommended that 2P be kept fully counterclockwise except in those

applications which specially require a narrow reference deadband between forward and reverse.

GAIN ADJUSTMENT

Turning 5P clockwise increases the gain of the drive system. High gain makes the system very responsive to transient changes in the motor load or the reference signal. As in any closed loop feedback system, high gain can contribute to system instability as well as transient response overshoot and ringing. Therefore, 5P is factory-set to minimum gain (full CCW) and it is recommended that 5P be kept in the minimum gain position except in those applications which specifically require high gain.

BALANCE ADJUSTMENT

Balance Potentiometer 7P is used to establish the reference voltage which will stop the motor. 7P is factory-adjusted by setting the reference signal to zero (jumpering 2TB-B to 2TB-C) and adjusting 7P until the motor stops. This is the proper setting for most applications.

In some applications it may be desirable to have the motor stop at some reference voltage other than zero. For these applications, set the reference signal to voltage at which it is desired that the motor stop, and then turn 7P until the motor does stop.

MAXIMUM SPEED ADJUSTMENTS

Turning 1P clockwise will increase the maximum reverse speed at which the motor will run. Turning 6P clockwise will increase the maximum forward speed at which the motor will run. Electrically these potentiometers accomplish their function by limiting the range of the speed reference potentiometer in the control station.

Potentiometers 1P and 6P are factory-set to limit the motor speed to a value about 10% above the motor nameplate rating. The customer can adjust these potentiometers by turning the reference signal potentiometer in the control station to one extreme and adjusting the maximum speed, and then repeating the operation for the other extreme. It is recommended that motors never be operated at speeds above the nameplate rating.

MAINTENANCE

MOTOR INSPECTION

Limited life items in the motor include the brushes, the commutator, the bearings, and the winding insulation. If the motor is not overloaded, the bearings and insulation should provide years of troublefree operation. However, the brushes and the commutator will wear with use. Factors which shorten brush life include high speed operation, high armature current, high armature voltage, extremely low humidity, and scratches or pits on the commutator surface. In severe duty applications the brushes should be inspected after every 1000 hours of operation. The brushes should be replaced anytime it is found that less than 1/2 inch of brush material remains. The bearings are packed with a special grease during assembly and should not be relubricated.

Make sure that the motor mounting bolts are all tight. A loose mounting bolt can cause considerable noise and may result in excessive wear.

If the motor requires maintenance, please refer to the General Electric Motor Maintenance Manual for the particular motor used on the application.

CONTROLLER INSPECTION

Inspect the contacts of the M relay, and replace the relay if the contacts are worn. Keep the heat sink clean, and do not allow anything to block the flow of air across the heat sink and through the enclosure.

TROUBLESHOOTING AND REPAIR

Most minor operational problems can be traced to improper adjustment of the various potentiometers on the component board. Proper adjustments are described in the "Potentiometer Adjustments" section of this book. Other possible problems are discussed below.

WARNING

SINCE THE CONTROLLER STOP BUTTON DOES NOT REMOVE VOLTAGE FROM THE CONTROLLER OR THE MOTOR FIELD, A BRANCH CIRCUIT BREAKER OR A FUSED DISCONNECT MUST BE USED TO DISCONNECT THE CONTROLLER WHENEVER WORK IS TO BE PERFORMED ON THE DRIVE.

If the Motor Will Not Run:

1. Be sure all connections are correct and tight. Check fuse 1FU, 2FU, 3FU, 4FU, and 5FU.

2. Be sure there is AC power available at the controller terminal. Check power switch closure.

3. Check the power fuses 1FU and 2FU. If a fuse is blown, replace it. If it blows again look for short circuit or motor overload.

CAUTION

THESE FUSES MUST BE HIGH INTERRUPTING CAPACITY, CERAMIC BODY, 250 VOLT FUSES, 1/4 INCH IN DIAMETER AND 1-1/4 INCHES LONG. RECOMMENDED FUSES ARE LITTLE-FUSE 314000 SERIES, TYPE 3AB "NORMAL BLO" FUSES AND BUSSMANN TYPE ABC FUSES.

If new fuses also blow, refer to Table I and check the settings of 3P and 4P. With an analyzer set on the X1000 ohms scale, verify that the reading from the cathode to the anode of the SCR's is open circuit. If the reading is not open circuit, one or both of the SCR's may have failed short.

If the fuse is not blown, but the motor still will not run, check the field voltage from 1TB-F1 to 1TB-F2. The field voltage should be 200 volts DC. If there is no field voltage, check the field power supply diodes which are mounted on the power interface board.

4. Check the motor for internal shorts. An ohmmeter should read one to five ohms from A1 to A2, and should read from 100 to several hundred ohms from F1 to F2.

Rotating the armature slowly while reading the resistance helps check for intermittent shorts in the motor.

If the motor runs faster or slower than normal for a given setting of the speed control:

1. Check the line voltage to be sure that it is normal, and recheck fuse 4FU.

2. Check the reference voltage to be sure that it is normal. If it is low, check the setting of the max speed potentiometers 1P and 6P.

3. Check the current limit adjustment.

4. Check for an overload on the motor. The motor is overloaded if the armature current exceeds the motor nameplate rating.

If the motor runs at full speed and cannot be controlled by speed control:

1. Check all connections for poor connection or short circuits. Check fuse 3FU.

2. Disconnect line power and check the SCR's with an ohmmeter set to the X1000 scale. The SCR's should read open circuit from the cathode to the heat sink.

3. With the motor running, check the voltage from main printed circuit board terminal 13A to 2TB-C. This is the feedback signal and must be equal to armature voltage. If this voltage is not equal to armature voltage, disconnect the power and check continuity from point 13A to 1TB-A1.

If the motor will run in only one direction:

1. Verify that the reference voltage measured from 2TB-B to 2TB-C is correct. 2TB-B must be positive with respect to 2TB-C for reverse rotation and negative with respect to 2TB-C for forward rotation.

2. Firing circuit 2FC triggers 2 SCR to make the motor go forward. 1FC triggers 1 SCR to make the motor go in the reverse direction. If the motor will not run in one direction, replace the corresponding firing circuit component board. If this does not solve the problem, replace the corresponding SCR.

If all previous troubleshooting steps have been unsuccesful:

1. Check the voltage on the transformer leads where the leads connect to the component board. The voltages should be as shown in Table I. If any voltages are wrong disconnect the control from the power line and disconnect the questionable transformer leads from the component board. Isolate the loose leads from all conductors, open circuit the M relay, connect the control to the power line and check the voltage again. If the voltage is now correct, the main component board should be replaced.

2. Replace the main component board.

RENEWAL PARTS

When ordering renewal parts, contact the nearest General Electric Sales Office, giving the quantity required and catalog numbers. In addition, give the 6VHRS number and the complete nameplate rating of the control.

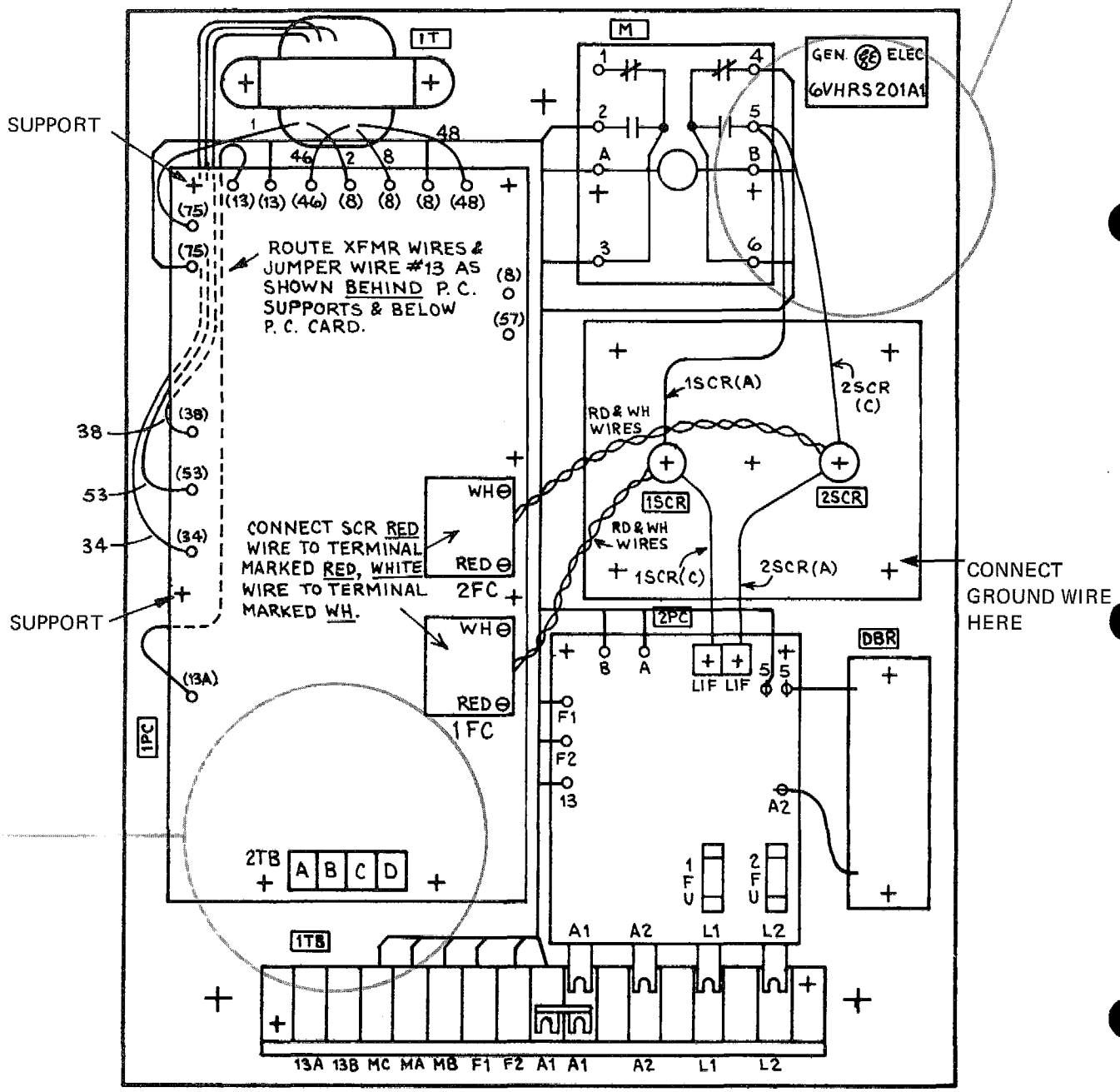
TABLE I
IT TRANSFORMER VOLTAGES

CIRCUIT NUMBERS	NOMINAL VOLTAGES
8 to 46	16 volts RMS ± 20%
8 to 48	16 volts RMS ± 20%
38 to 43	4 volts RMS ± 20%
38 to 53	4 volts RMS ± 20%

TABLE II

PART NOMENCLATURE	CATALOG NUMBER	QUANTITY REQUIRED
1PC	193X638AAG02	1
2PC	193X641AAG02	1
1, 2 SCR	36A358006AAG06	2
M RELAY	104X131AB001	1
1T	36A349214AAG02	1
3, 4, 5 FU	104X109AD028	3
1, 2 FU	104X109AD037	2
DBR	6VSJ03A1	1

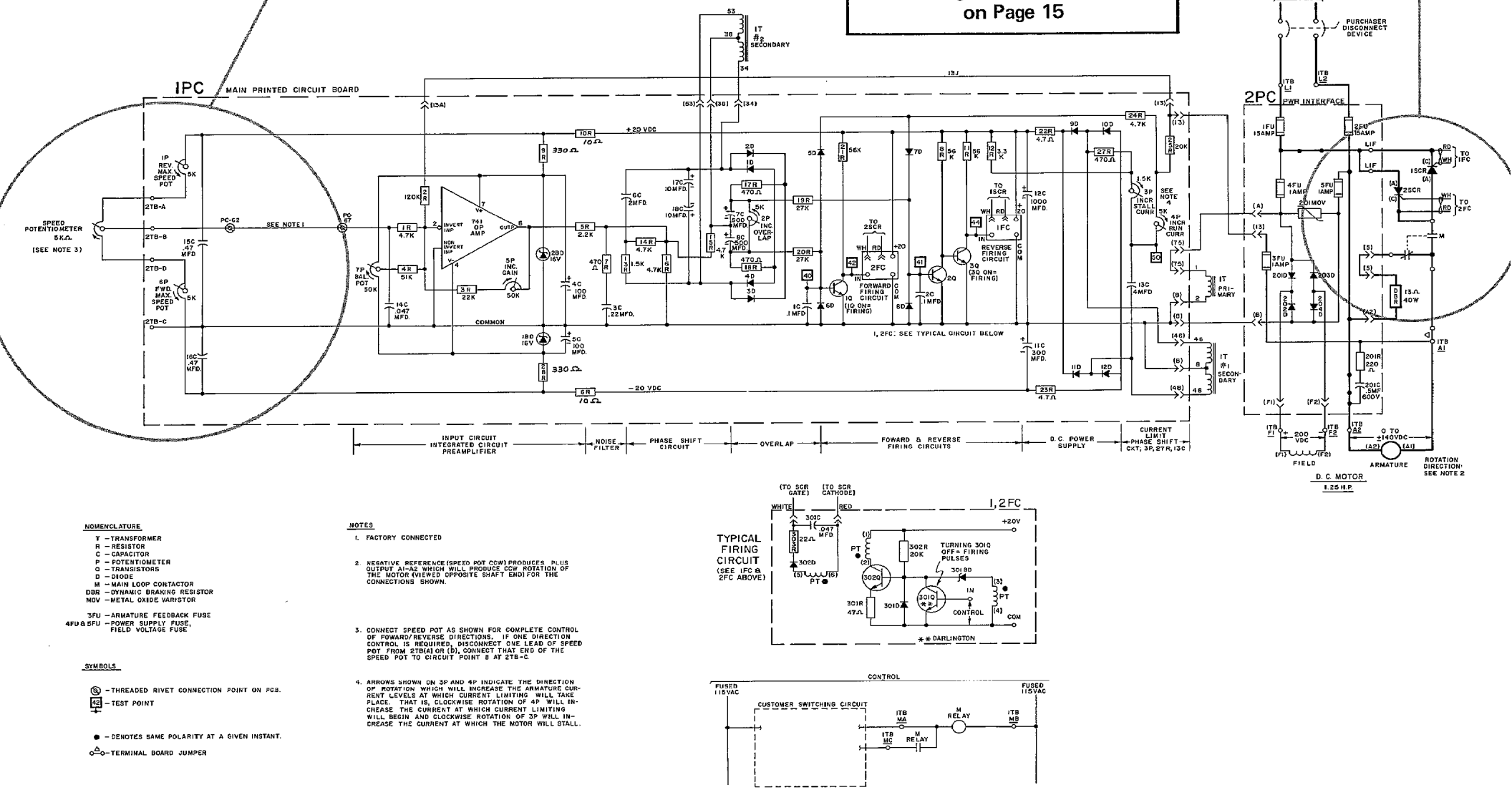
IMPORTANT
See Figure 4 Supplement
on Page 15



SCR LEAD INDENTIFICATION: 1, 2SCR(C) IS THE BLACK LEAD WHICH ALSO HAS A RED WIRE SOLDERED AT THE SCR. 1, 2SCR(A) IS THE OTHER BLACK WIRE.

Figure 4. 6VHRS Internal Connection Diagram

IMPORTANT
See Figure 5 Supplement
on Page 15



NOMENCLATURE

T – TRANSFORMER
R – RESISTOR
C – CAPACITOR
P – POTENTIOMETER
Q – TRANSISTORS
D – DIODE
M – MAIN LOOP CONTACTOR
DBR – DYNAMIC BRAKING RESISTOR
MOV – METAL OXIDE VARISTOR

3FU – ARMATURE FEEDBACK FUSE
4FU & 5FU – POWER SUPPLY FUSE, FIELD VOLTAGE FUSE

SYMBOLS

Ⓢ – THREADED RIVET CONNECTION POINT ON PCB.
42 – TEST POINT
● – DENOTES SAME POLARITY AT A GIVEN INSTANT.
– TERMINAL BOARD JUMPER

NOTES

1. FACTORY CONNECTED
2. NEGATIVE REFERENCE (SPEED POT CCW) PRODUCES PLUS OUTPUT AT A2 WHICH WILL PRODUCE CCW ROTATION OF THE MOTOR (VIEWED OPPOSITE SHAFT END) FOR THE CONNECTIONS SHOWN.
3. CONNECT SPEED POT AS SHOWN FOR COMPLETE CONTROL OF FOWARD/REVERSE DIRECTIONS. IF ONE DIRECTION CONTROL IS REQUIRED, DISCONNECT ONE LEAD OF SPEED POT FROM 2TB(A) OR (D), CONNECT THAT END OF THE SPEED POT TO CIRCUIT POINT 8 AT 2TB-C.
4. ARROWS SHOWN ON 3P AND 4P INDICATE THE DIRECTION OF ROTATION WHICH WILL INCREASE THE ARMATURE CURRENT LEVELS AT WHICH CURRENT LIMITING WILL TAKE PLACE. THAT IS, CLOCKWISE ROTATION OF 4P WILL INCREASE THE CURRENT AT WHICH CURRENT LIMITING WILL BEGIN AND CLOCKWISE ROTATION OF 3P WILL INCREASE THE CURRENT AT WHICH THE MOTOR WILL STALL.

Figure 5. 6VHRS201A1 Schematic Diagram

FRONT VIEW
SHOWING LOCATION OF COMPONENTS

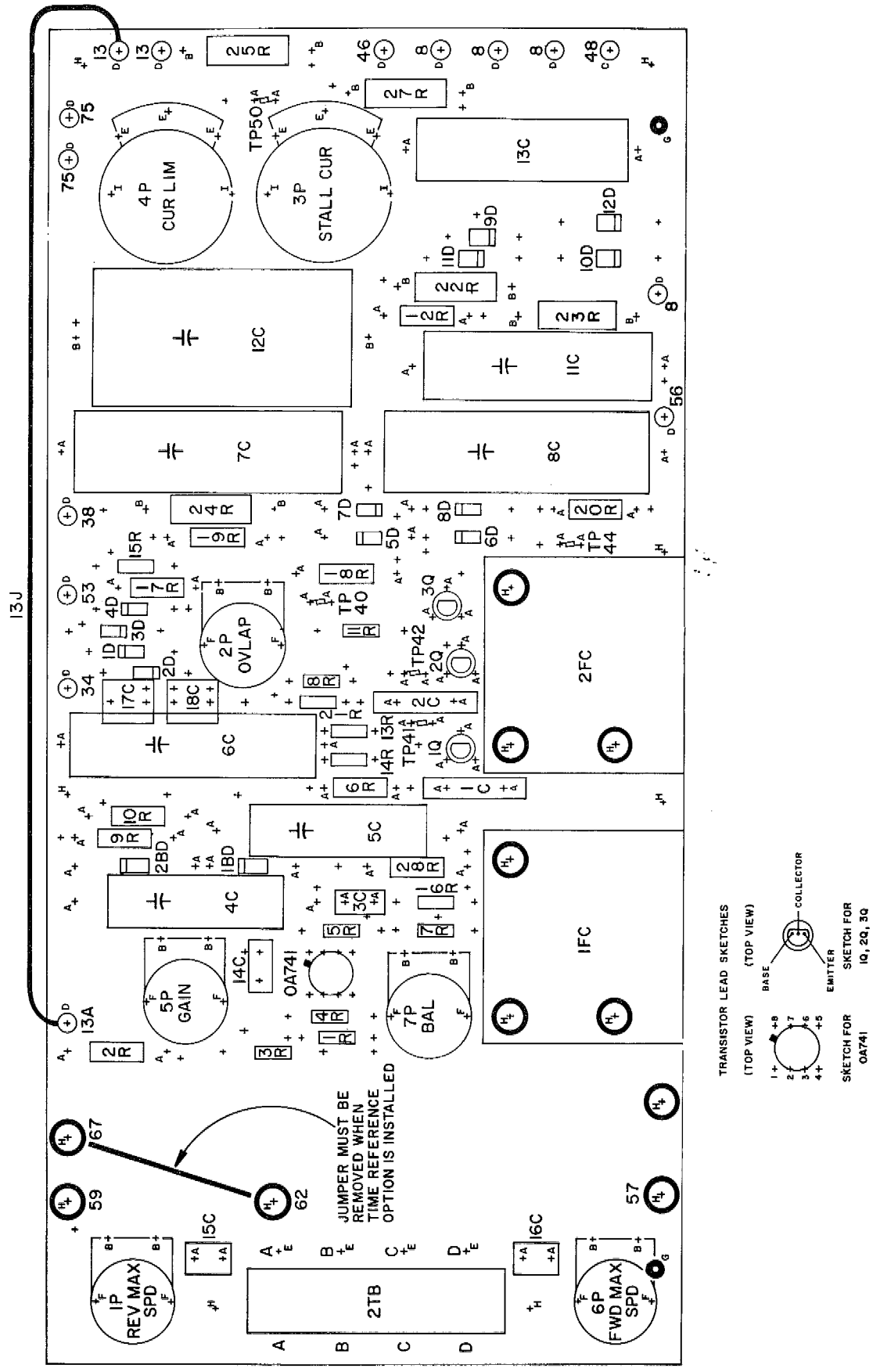
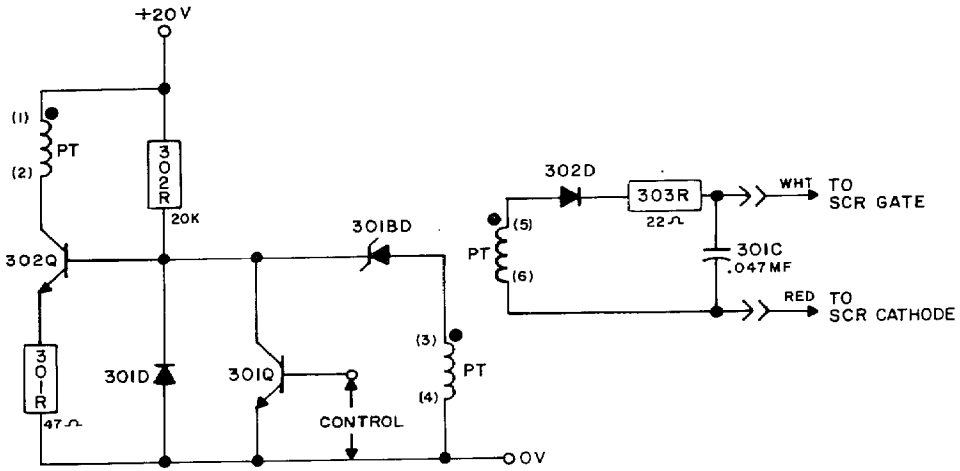
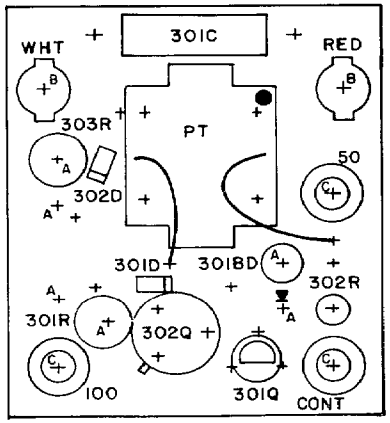


Figure 6. Component Diagram for Main Board

SCHEMATIC DIAGRAM



FRONT VIEW
SHOWING LOCATION OF COMPONENTS



193X640AAG01
FIRING CIRCUIT

TRANSISTOR LEAD SKETCHES

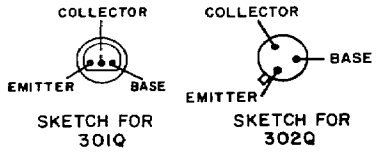


Figure 7. Component Diagram for Firing Circuit Board

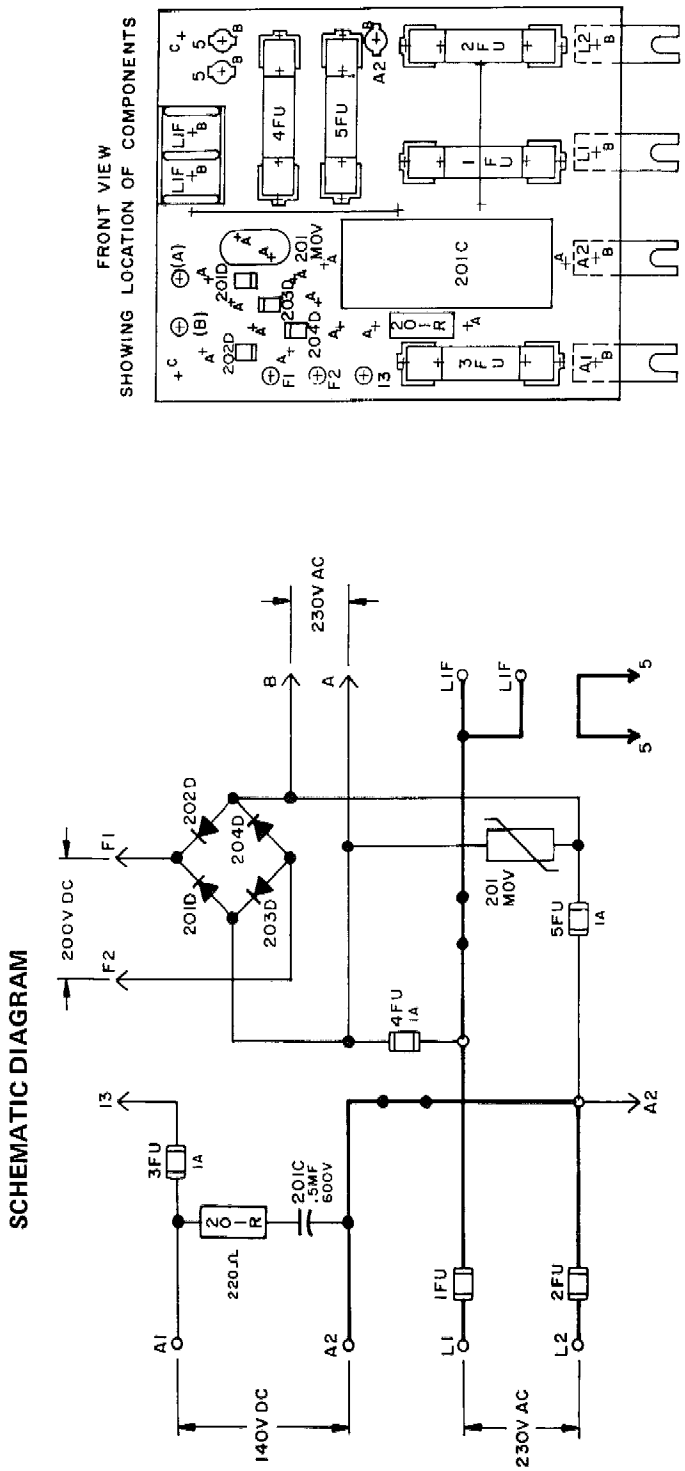


Figure 8. Component Diagram for Power Interface Board

— NOTES —

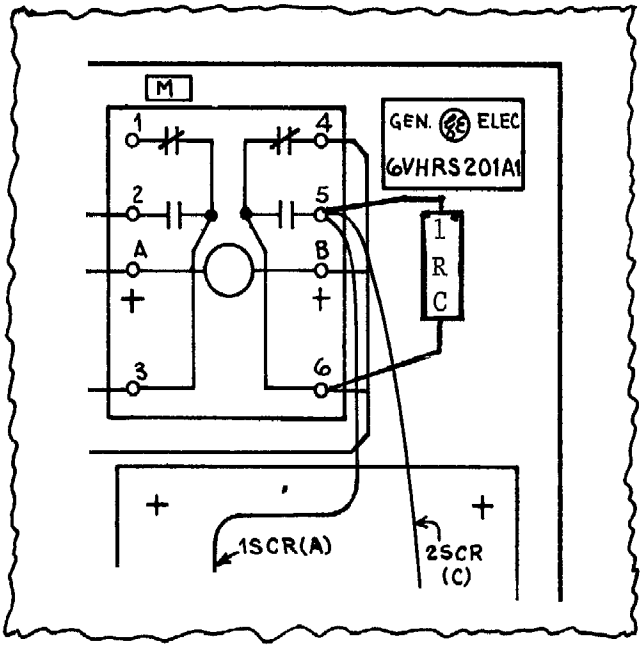
GEK-47613 SUPPLEMENT

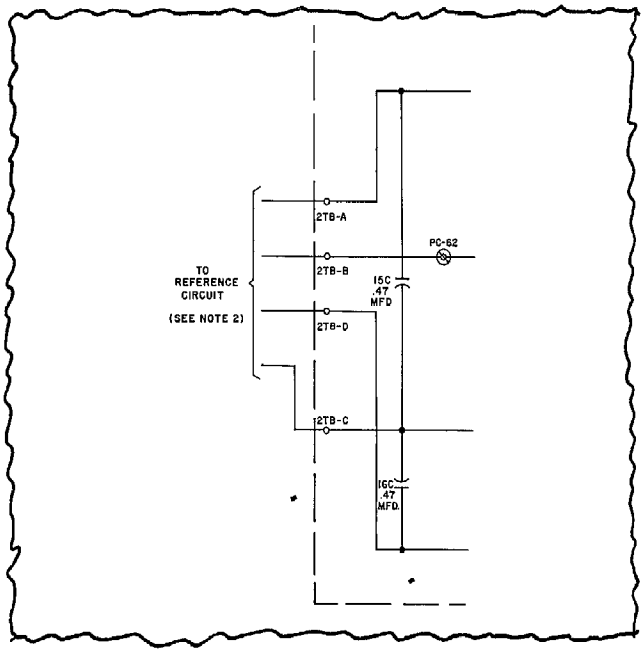
THIS SUPPLEMENT CONTAINS INFORMATION PERTAINING TO CIRCUIT CHANGES WHICH HAVE BEEN INCORPORATED IN THE 6VHRS201A1 STATOTROL CONTROLLER. THE CHANGES ARE:

1. Addition of an R-C circuit (1RC).

2. Deletion of maximum speed potentiometers from the main printed circuit board.

FIGURE 4 Supplement and FIGURE 5 Supplement depict a partial connection and partial schematic diagram for the addition of 1RC and the substitution of wire jumpers in place of maximum speed potentiometers.





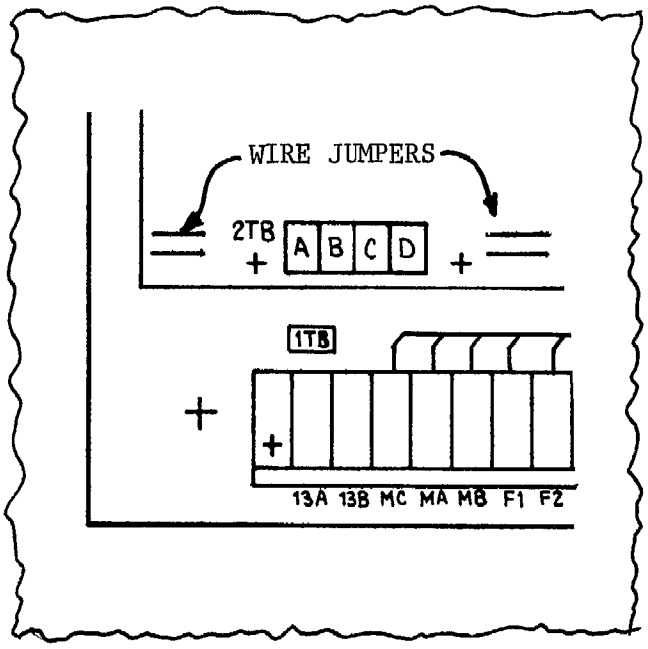


Figure 4 Supplement

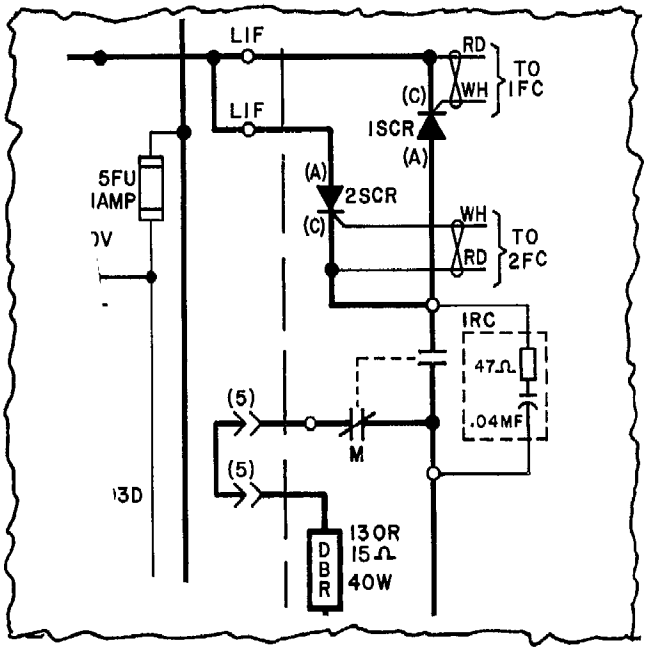


Figure 5 Supplement

GENERAL ELECTRIC COMPANY – DIRECT CURRENT MOTOR & GENERATOR DEPARTMENT
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC

GEK-47613 (11/75) 1.5M D